Project Report On

"Smart Waste Management System"

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DECLARATION

I hereby declare that the synopsis report entitled "Smart Waste And Management" by Aditi Dobhal to Uttaranchal School of Computing Sciences, Uttaranchal University the project was done under the guidance of Ms. Divya Rawat. I further declare that the work reported in this project has not been submitted and will not been submitted, either in part or in full for the award of any other degree or diploma in this university or any other university or institute.

CERTIFICATE OF ORIGINALITY

This is to certify that the project report entitled "smart waste management system" submitted by Sakshi Rawat is an original work carried out by the student under proper guidance as a partial fulfilment of the requirements for the Mini Project of Bachelor of Computer Applications (BCA) at School of Computing Sciences (USCS), Uttaranchal University.The work presented in this report has been completed by the student through his own efforts, learning, and understanding. To the best of our knowledge, this project has not been submitted earlier, either in part or in full, to any other university or institution for the award of any degree or diploma.

Under the guidance of:

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1.INTRODUCTION

In recent years, rapid urbanization and population growth have significantly strained traditional waste management systems, leading to inefficiencies such as overflowing bins, high operational costs, and increased environmental pollution. Conventional waste collection typically follows fixed, periodic schedules regardless of whether a bin is actually full, resulting in unnecessary fuel consumption and labour. The IoT-based Smart Waste Management System addresses these challenges by integrating hardware, such as ultrasonic sensors and microcontrollers, with cloud-based data analytics to enable a real-time, data-driven approach to urban sanitation. Core Concepts of the Project Real-Time Monitoring: Utilizing ultrasonic sensors to measure the distance between the bin lid and the garbage, the system calculates the "fill level" and transmits this data to a central cloud platform. Dynamic Route Optimization: Instead of following static routes, collection vehicles are dispatched only to bins that have reached a predefined threshold (e.g., 80% full). This can reduce collection mileage by up to 20% and lower operational costs by 30% Automated Interaction: Many smart bins incorporate servo motors and motion sensors to provide a contactless, hygienic experience for users, which has become a priority in post-pandemic urban design. Advanced Features: Beyond level monitoring, modern prototypes often include sensors for weight temperature/humidity , and even waste segregation (wet vs. dry) to prevent hazardous decomposition and improve recycling rates.

2.OBJECTIVES

- ## To Optimize Waste Disposal Accessibility

  To develop a system that allows users to instantly locate the nearest dustbins based on their current coordinates and specific waste category (Dry or Wet), reducing littering caused by the inability to find disposal points.

- ## To Enhance Urban Hygiene through Data-Driven Mapping

  To create a digital repository of waste disposal points that calculates and provides the most efficient routes for users, supporting Smart City initiatives and cleanliness missions like Swachh Bharat.

3.SYSTEM ANALYSIS

1. Input Phase

In this project, the user provides information that helps the system find nearby dustbins.

Inputs include:

- User’s current location or area name
- Type of waste (dry waste or wet waste)
- Dustbin location data stored in the system database
- Distance or coordinates of dustbins

Example:

A user enters their area name or location on the website and selects whether they want to dispose dry waste or wet waste.

2. Logic Phase

The website compares the user’s location with the dustbin locations stored in the system and identifies the nearest dustbin.

Logic used in the system:

- Compare user location with stored dustbin locations
- Check the type of dustbin (dry or wet)
- Filter the dustbins based on the user’s requirement
- Select the nearest dustbin

Example:

If a user selects wet waste, the system will only search for wet waste dustbins near the user’s location.

3.Output Phase

Outputs include:

- Displaying the nearest dustbin location
- Showing whether it is for dry waste or wet waste
- Showing the distance from the user
- Displaying the dustbin location on the website

4. IDENTIFICATION OF NEED

In today's world, due to increasing population and lack of civic sense, you can find garbage in literally any part of the country.

This is very concerning and makes the need for this project very important.

In many places people often face difficulty in finding nearby dustbins while travelling , shopping , or visiting public places. Due this many people throe garbage on roads , parks , or in open area , which causes environmental pollution and unhygienic conditions .

- **Problems in Existing System**

1. No online platform to check nearby dustbin locations.
2. Lack of awareness about separate bins for wet and dry waste.
3. Difficulty in locating dustbins in emergency situations.
4. Increased littering due to unavailability of information.

- **Need for the System**

1. Detect and display nearby dustbins using location services.
2. Show separate dustbins for wet and dry waste.
3. Provide distance from user’s current location.
4. Help users navigate easily to the nearest dustbin.
5. Promote cleanliness under initiatives like Swachh Bharat Mission.

- **Such a system will:**

1. Reduce littering
2. Improve public hygiene
3. Support smart city development
4. Encourage responsible waste disposal

5.PRELIMINARY INVESTIGATION

Before starting the development of this project, a preliminary investigation was conducted to understand the current problems faced in waste management.

It was observed that people often do not know the location of nearby dustbins. In many areas, there is no proper system to guide users toward the correct dustbin according to waste type.

After analysing these problems, it was concluded that a smart system is required that can:

- Store dustbin information
- Calculate nearest dustbin location
- Differentiate between dry and wet waste bins
- Provide an easy-to-use interface

This investigation helped in planning and designing the proposed system effectively.

6. FEASIBILITY ANALYSIS

Technical

The project is technically feasible because it can be developed using basic programming knowledge of C language and web technologies.

Economic

The system does not require expensive hardware or software, so it is cost-effective.

Operational

The website will be easy to use for both users and administrators, making it practical for daily use.

7. PROJECT PLANNING

The project was planned in the following stages:

1. Requirement Collection
2. System Analysis
3. System Design
4. Coding and Implementation
5. Testing
6. Documentation

Each stage was completed step by step to ensure proper project development

8.SYSTEM REQUIREMENTS

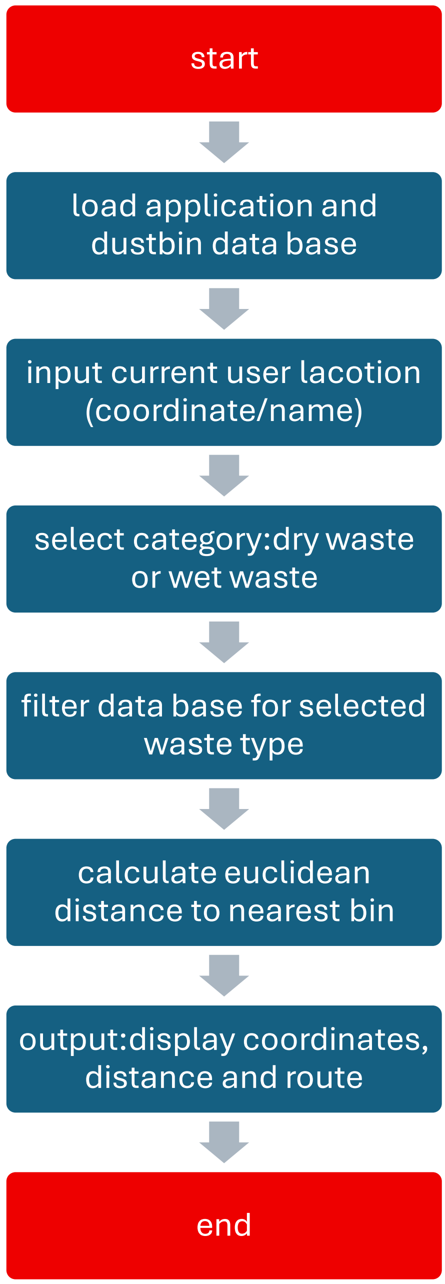
Software:

- C Language
- GCC / Turbo C Compiler
- Code::Blocks / Dev C++
- Windows OS

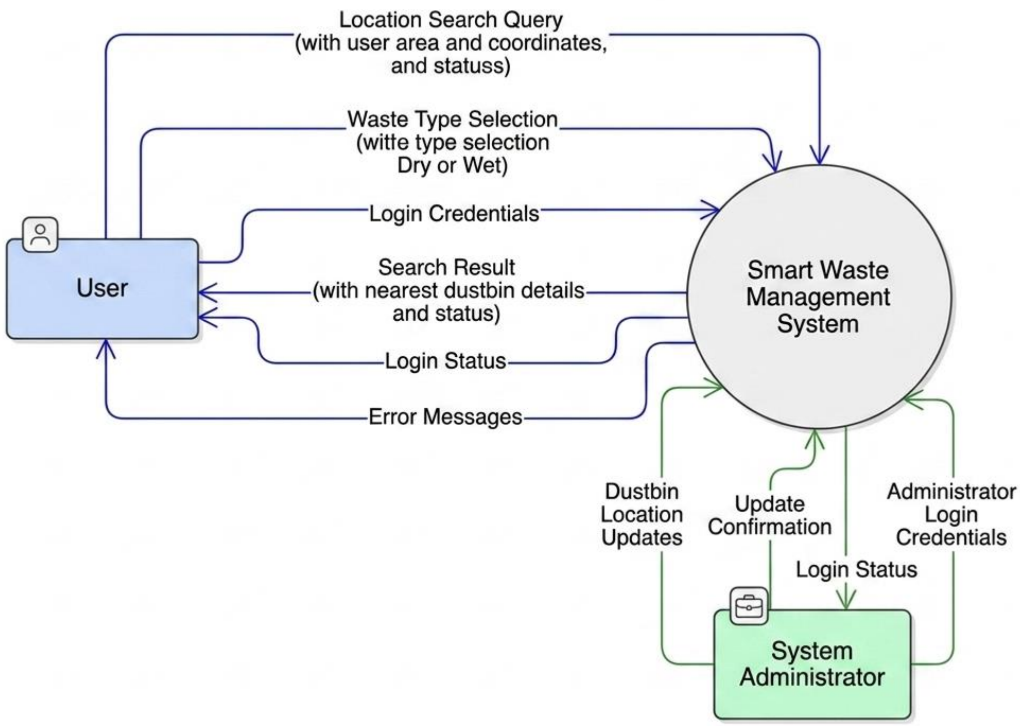
Hardware:

- Computer / Laptop
- 4 GB RAM
- 100 GB Storage
- Keyboard and Mouse

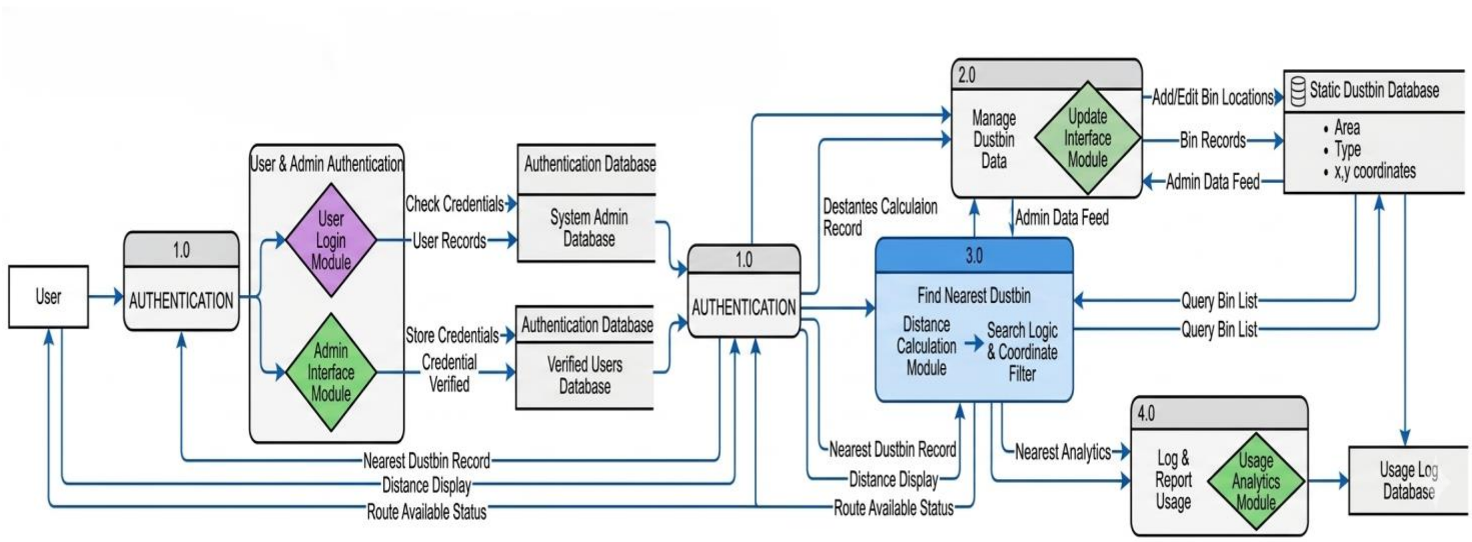
9.FLOW CHART


start
load application and dustbin data base
input current user lacotion (coordinate/name)
select category:dry waste or wet waste
filter data base for selected waste type
calculate euclidean distance to nearest bin
output:display coordinates, distance and route
end

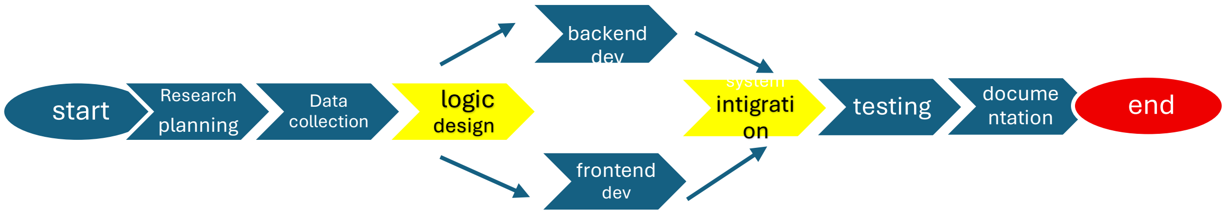
10. LEVEL 0 DATA FLOW DIAGRAM



11. LEVEL 1 DATA FLOW DIAGRAM



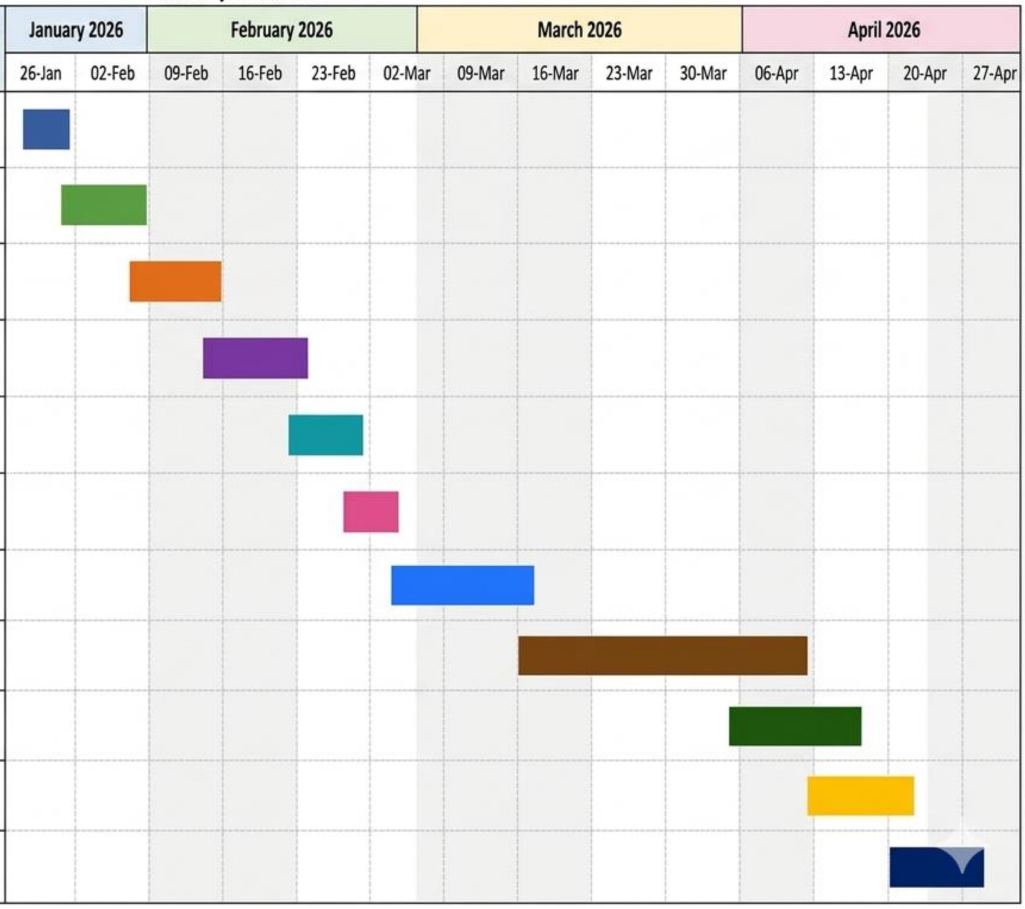
12. Pert chart



13.Gantt chart

Mini Project Schedule

No.	Task	Start Date	End Date	Duration (Days)
1	Identification of Need	27-Jan-2026	31-Jan-2026	4 Days
2	Preliminary Investigation	31-Jan-2026	06-Feb-2026	6 Days
3	Feasibility Study	06-Feb-2026	13-Feb-2026	7 Days
4	System Analysis	13-Feb-2026	21-Feb-2026	8 Days
5	Project Planning	21-Feb-2026	27-Feb-2026	6 Days
6	Project Scheduling	27-Feb-2026	03-Mar-2026	4 Days
7	System Design	03-Mar-2026	15-Mar-2026	12 Days
8	Coding & Development	15-Mar-2026	04-Apr-2026	20 Days
9	Testing	04-Apr-2026	16-Apr-2026	12 Days
10	Implementation	16-Apr-2026	23-Apr-2026	7 Days
11	Documentation	23-Apr-2026	30-Apr-2026	7 Days



11.SYSTEM DESIGN

1 Module Details

Login Module

- Handles user authentication
- Verifies username and password

User Input Module

- Accepts user coordinates
- Accepts waste type selection

Dustbin Management Module

- Stores dustbin details
- Maintains dry and wet waste categories

Distance Calculation Module

- Calculates shortest distance using Euclidean formula

Output Module

- Displays nearest dustbin details
- Shows distance and coordinates

2 Data Integrity and Constraints

- Coordinates must be numeric

- Waste type must be Dry or Wet
- Invalid choices are restricted
- Dustbin data should remain properly stored

3 Structure Design

```
struct Dustbin
{
    char area[30];
    char type[10];
    int x, y;
};
```

4 User Interface Design

The system uses a console-based menu-driven interface.

Main Features

- User Login
- Waste Type Selection
- Nearest Dustbin Search
- Distance Display

5 System Constraints

- Console-based interface only
- No GPS integration
- Static dustbin data
- No database connectivity

12.CODE

```
#include <stdio.h>
#include <string.h>
#include <math.h>

struct Dustbin
{
    char area[30];
    char type[10];
    int x, y;
};

int main()
{
    char username[30], password[20];
    char name[30], waste[10];
    int ux, uy, i, choice;
    int index = -1;
    float dist, min = 9999;

    struct Dustbin d[8] =
```

```
{
    {"Clock Tower", "Dry", 2, 3},
    {"Rajpur Road", "Wet", 5, 6},
    {"ISBT", "Dry", 8, 2},
    {"Prem Nagar", "Wet", 1, 7},
    {"Ballupur", "Dry", 6, 4},
    {"Saharanpur Chowk", "Wet", 3, 8},
    {"Jakhan", "Dry", 7, 6},
    {"Patel Nagar", "Wet", 4, 2}
};

/* LOGIN PHASE */
printf("=====================================\n");
printf(" SMART WASTE MANAGEMENT SYSTEM\n");
printf("=====================================\n");

printf("\n---------- LOGIN ----------\n");
printf("Enter Username : ");
scanf("%s", username);

printf("Enter Password : ");
scanf("%s", password);

if(strcmp(username, "admin") == 0 && strcmp(password, "1234") == 0)
{
   printf("\nLogin Successful!\n");
}
else
```

```
{
    printf("\nInvalid Username or Password!\n");
    return 0;
}

/* USER DETAILS */
printf("\n---------- USER DETAILS ----------\n");
printf("Enter Your Name : ");
scanf("%s", name);

printf("Enter Your Current Location Coordinates (x y): ");
scanf("%d%d", &ux, &uy);

printf("\nSelect Waste Type:\n");
printf("1. Dry Waste\n");
printf("2. Wet Waste\n");
printf("Enter Choice : ");
scanf("%d", &choice);

if(choice == 1)
    strcpy(waste, "Dry");
else if(choice == 2)
    strcpy(waste, "Wet");
else
{
    printf("Invalid Choice!\n");
    return 0;
}
```

```
/* DISTANCE CALCULATION */
for(i = 0; i < 8; i++)
{
    if(strcmp(d[i].type, waste) == 0)
    {
        dist = sqrt(pow(ux - d[i].x, 2) + pow(uy - d[i].y, 2));

        if(dist < min)
        {
            min = dist;
            index = i;
        }
    }
}

/* OUTPUT */
printf("\n====================================\n");
printf(" SEARCH RESULT\n");
printf("====================================\n");

printf("User Name      : %s\n", name);
printf("Waste Type     : %s\n", waste);

if(index != -1)
{
    printf("Nearest Dustbin: %s\n", d[index].area);
    printf("Coordinates    : (%d,%d)\n", d[index].x, d[index].y);
```

```c
        printf("Distance        : %.2f KM\n", min);

        printf("Status          : Route Available\n");

    }

    else

    {

        printf("No Dustbin Found!\n");

    }

      printf("\nThank You For Keeping Dehradun Clean!\n");

return 0;}
```

13. TESTING

```
=====================================
 SMART WASTE MANAGEMENT SYSTEM
=====================================

---------- LOGIN ----------
Enter Username : admin
Enter Password : 1234

Login Successful!

---------- USER DETAILS ----------
Enter Your Name : aditi
Enter Your Current Location Coordinates (x y): 2 3

Select Waste Type:
1. Dry Waste
2. Wet Waste
Enter Choice : 1

=====================================
 SEARCH RESULT
=====================================
User Name       : aditi
Waste Type      : Dry
Nearest Dustbin: Clock Tower
Coordinates     : (2,3)
Distance        : 0.00 KM
Status          : Route Available

Thank You For Keeping Dehradun Clean!

--------------------------------
Process exited after 14.38 seconds with return value 0
Press any key to continue . . .
```

Testing Techniques Used

1. Black Box Testing

- Tested login validation
- Tested waste type selection
- Tested coordinate inputs

2. White Box Testing

- Verified loops and conditions
- Checked distance calculation logic

3. Unit Testing

Modules tested individually:

- Login Module
- Input Module
- Distance Calculation Module
- Output Module

4. System Testing

Complete system workflow was tested successfully.

14. System Security Measures

Authentication Security

- Username and password verification implemented
- Unauthorized users restricted

Input Validation

- Invalid menu choices handled
- Proper waste type validation applied

Error Handling

- Wrong login credentials handled safely
- Incorrect inputs display error messages

Limitations

- No password encryption
- No database security
- Console-based security only

15. FUTURE SCOPE

1. IoT Integration

Install ultrasonic sensors in physical dustbins to provide real-time fill levels. This would prevent users from navigating to a bin that is already full.

2. Live Map Navigation

Integrate the Google Maps API to replace static coordinates with live, turn-by-turn navigation and global location support.

3. Reward System

Introduce a QR-code based incentive program where users earn points for proper waste disposal, which can be redeemed for local coupons or digital rewards.

4. AI Image Recognition

Use Machine Learning to allow users to scan an item with their camera; the app will then automatically identify if it is "Dry" or "Wet" waste.

5. Mobile App Development

Transition the current web-based logic into a dedicated Android and iOS application for easier access and push notifications for nearby bins.

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